

BRAC University
Department of Electrical and Electronic Engineering
EEE 221L: Energy Conversion Laboratory
Fall 2025
Project

Marks: 20

Problem:

You are required to construct/build and analyze a single-phase tap-changing transformer. The project includes detailed calculations, core/winding construction, performance testing, cost estimation, and justification of tap-changing methods.

Design Parameters:

1. Primary Voltage: X-220-A-0
 - a. X = 5% above 220 V
 - b. A = 5% below 220 V
2. Secondary Voltage: 24 V
3. Frequency: 50 Hz
4. Primary Current Rating: Select as required
5. Core Material: Laminated sheet iron
6. Tap Positions: Two primary taps

Tasks:

1. Provide a complete step-by-step design explanation.
2. Show all calculations with reference documents (self study); such as:
 - a. Core area
 - b. Turns per volt
 - c. Primary & secondary turns
 - d. Wire gauge
 - e. Copper loss, iron loss, efficiency
3. Construct the transformer.
4. Perform tests:
 - a. Polarity test
 - b. No-load test
 - c. Load test and regulation
 - d. Tap voltage verification
 - e. Efficiency
5. Compare results with theoretical values.
6. Prepare a complete budget and compare with actual cost.

Equipment: Purchase from the local market and share the cost within the group members.

1. Laminated iron core
2. Lamination paper
3. Copper wire
4. Insulation materials
5. Electrical connectors
6. Voltmeter, ammeter, wattmeter
7. Resistive load bank
8. Workshop tools

Instructions:

1. **Each group should share the cost among the group members.**
2. Take a slot from the **Lab Officer** to perform the necessary tests of your constructed Transformer with your group in the laboratory. The deadline to collect data in the lab is **31st December, 2025. After that, you won't get any support from the lab.**
3. You must prepare a detailed report based on the construction, collected data and the calculations.
4. **In the report**, you need to add a detailed description of the transformer, construction and testing procedures, theory, circuits, types of tests you performed, loading effect, a description of tools, techniques, or methods used, modifications, meter ranges, practices, issues, troubleshooting, error percentage, equivalent circuit parameters, budget etc.
5. During the test, you should **present in the lab**. The performance will be observed. Take a signature after collecting data from the lab officer mentioning your name and ID on the paper and submit it with the report.
6. Take the necessary photos to include as an appendix to your report.
7. Submit the **handwritten report with a given cover page** on or before **January 7, 2026. This is a group submission. Your report should not be identical to the other groups.**
8. This project will be assessed for Course outcome.

Course Outcome (CO) Description Table

Sl.	CO Description	POs	Bloom's Taxonomy Domain/Level
CO5	Demonstrate proficiency in using laboratory tools to carry out experiments.	e	Cognitive/Understand, Psychomotor/Precision

(e) Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering & IT tools, including prediction and modeling, to complex electrical and electronic or electronic and communication engineering problems, with an understanding of their limitations. (K6)	e1. Select appropriate techniques, resources, modern engineering, and IT tools to solve complex engineering problems. e3. Use appropriate modern engineering, IT including modelling and simulation tools and techniques to develop and evaluate the solution of complex engineering problems
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Rubrics for Assessment

Description	e1: Select appropriate techniques, resources, modern engineering, and IT tools to solve complex engineering problems e3. Use appropriate modern engineering, IT including modelling and simulation tools and techniques to develop and evaluate the solution of complex engineering problems	
Performance Level	Selection of Tools, Techniques, Advantages, and Limitations (10 Marks)	Issues, Troubleshooting, Result Analysis (10 Marks)
Excellent	10	10
Satisfactory	8	8
Somewhat Satisfactory	6	6
Unsatisfactory	4	4

Performance Criteria	Performance Measure			
	Unsatisfactory (2)	Developing (5)	Satisfactory (8)	Exemplary (10)
e1 Select appropriate tools [10]	Does not have basic knowledge of using any tools. Need significant help from others to finish the assigned work	Learned a minimum level of technique just to get the project done. Need time-to-time help from others	Knows more than one technique and able to complete the assigned work without help	Knows several techniques to solve the problem and can use the better one.
e3 Use appropriate engineering tools and techniques to construct	Not able to apply the modern engineering tools	Limited capability of applying modern engineering tools to develop the solution	Able to apply modern engineering tools satisfactorily to develop and evaluate	Demonstrate a complete understanding and organized manner of using modern engineering tools

the proper setup [10]			the solution. Familiar with the design process used by the modern tool.	to develop and evaluate the solution
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